

Experiment7

- 1) Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int global_var = 10; // Global variable

void code_function()
{
    printf("Code address : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20; // Static variable
    int stack_var = 30;        // Stack variable
    int *heap_var = malloc(sizeof(int)); // Heap variable

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    *heap_var = 40;

    printf("Linux Process Address Space\n");
    printf("-----\n");

    code_function();

    printf("Global address : %p\n", (void *)&global_var);
```

```

printf("Static address : %p\n", (void *)&static_var);
printf("Heap address   : %p\n", (void *)&heap_var);
printf("Stack address  : %p\n", (void *)&stack_var);

free(heap_var);

return 0;
}

```

OUTPUT :

```

ashwithreddy_earla@AshwithReddy:~$ nano Experiment8.c
ashwithreddy_earla@AshwithReddy:~$ nano Experiment8.c
ashwithreddy_earla@AshwithReddy:~$ gcc Experiment8.c -o Experiment8
ashwithreddy_earla@AshwithReddy:~$ ./Experiment8
Linux Process Address Space
-----
Code address   : 0x6359f80391c9
Global address : 0x6359f803c010
Static address : 0x6359f803c014
Heap address   : 0x635a12f72010
Stack address  : 0x7fffd9949450c

```

2) Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

int global_var = 100;

int main()
{

```

```
static int static_var = 200;

int stack_var = 300;

int *heap_var = malloc(sizeof(int));

*heap_var = 400;

printf("Process ID (PID): %d\n", getpid());

printf("Global variable address : %p\n", (void *)&global_var);

printf("Static variable address : %p\n", (void *)&static_var);

printf("Heap variable address : %p\n", (void *)heap_var);

printf("Stack variable address : %p\n", (void *)&stack_var);

printf("\nProcess is running...\n");

getchar();

free(heap_var);

return 0;

}
```

OUTPUT:

```
ashwithreddy_earla@AshwithReddy:~$ gcc Experiment7.1.c -o Experiment7.1
ashwithreddy_earla@AshwithReddy:~$ ./Experiment7.1
Process ID (PID): 710
Global variable address : 0x5f592b8f2010
Static variable address : 0x5f592b8f2014
Heap variable address : 0x5f59398b9010
Stack variable address : 0x7ffe7e69f2ac
```